An anisotropic obstruction to every $q > p$ reverse affine Zhang improvement

Candidate counterexample for arXiv:2003.07391

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Status. Candidate counterexample, likely valid. It gives a complete negative answer to Section 7, item (6) of the source. The other open problems in that section are not addressed.

1 Source question

Haddad, Jiménez, and Montenegro prove on a bounded open set $\Omega \subset \mathbb{R}^n$ the reverse affine Zhang inequality

$$\mathcal{E}_p f \geq C_{n,p}(\Omega)\|f\|_p^{(n-1)/n}\|\nabla f\|_p^{1/n}.$$

In Section 7, item (6), PDF page 27 of [1], they ask whether this can be improved by replacing $\|f\|_p$ with $\|f\|_q$ for some $q > p$.

> (6) We ask whether inequality (11) can be improved to
>
> $$\mathcal{E}_p f \geq C_{n,p}(\Omega)\|f\|_q^{\frac{n-1}{n}}\||\nabla f|\|_p^{1/n}$$
>
> for some parameter $q > p$.
>
> A positive answer would allow to prove existence of minimizers and compactness to mixed variational problems. Some work has been done in [57] in this direction for the case $p = 2$ and $2 < q < n$, suggesting that the previous inequality could be valid.

2 Result

Theorem 1. *Let $n \geq 2$, $p \geq 1$, $q > p$, and let $\Omega \subset \mathbb{R}^n$ be any nonempty open set. Then*

$$\inf_{0 \neq f \in C_c^\infty(\Omega)} \frac{\mathcal{E}_p f}{\|f\|_q^{(n-1)/n}\|\nabla f\|_p^{1/n}} = 0.$$

Consequently there is no positive constant, even one depending on n, p, q, and Ω, for the proposed $q > p$ inequality.

This shows that the exponent $q = p$ in the source theorem is sharp against replacement by every strictly larger Lebesgue exponent.

3 Proof intuition

Affine energy is designed to tolerate volume-preserving anisotropy. Compress a fixed bump in only one coordinate. Its ordinary gradient sees the inverse thickness ε^{-1}, while its affine energy sees only the determinant of the compression. The L^q norm contributes a different Jacobian power. After inserting all three exact scalings, the proposed ratio contains the positive power $\varepsilon^{\frac{n-1}{n}(1/p-1/q)}$. It therefore vanishes for every $q > p$. The construction is smooth and stays inside the original fixed domain.

4 Proof

Choose $x_0 \in \Omega$ and a rectangular neighborhood $x_0 + (-r,r)^n \subset \Omega$. Fix a nonzero $g \in C_c^\infty((-r,r)^n)$ such that $\partial_1 g \not\equiv 0$. For $0 < \varepsilon \le 1$, let

$$A_\varepsilon = \operatorname{diag}(\varepsilon, 1, \dots, 1), \qquad f_\varepsilon(x) = g\big(A_\varepsilon^{-1}(x - x_0)\big).$$

Then $f_\varepsilon \in C_c^\infty(\Omega)$.

We first derive the affine-energy scaling. For a smooth f, set

$$N_{f,p}(\xi) = \|\partial_\xi f\|_p, \qquad L_{f,p} = \{\xi \in \mathbb{R}^n : N_{f,p}(\xi) \le 1\}.$$

The source's geometric formula is $\mathcal{E}_p f = c_{n,p} n^{-1/n} |L_{f,p}|^{-1/n}$. For an invertible matrix A and $f_A(x) = f(A^{-1}x)$, the chain rule and change of variables give

$$\begin{aligned} N_{f_A,p}(\xi)^p &= \int_{\mathbb{R}^n} |\langle A^{-\mathsf{T}} \nabla f(A^{-1}x), \xi\rangle|^p \, dx \\ &= |\det A| \, N_{f,p}(A^{-1}\xi)^p. \end{aligned}$$

Therefore

$$L_{f_A,p} = A\big(|\det A|^{-1/p} L_{f,p}\big), \qquad |L_{f_A,p}| = |\det A|^{1-n/p} |L_{f,p}|,$$

and hence the exact covariance law is

$$\mathcal{E}_p(f_A) = |\det A|^{1/p - 1/n} \mathcal{E}_p(f). \tag{1}$$

Translation does not change any of the quantities. Since $\det A_\varepsilon = \varepsilon$, (1) yields

$$\mathcal{E}_p f_\varepsilon = \varepsilon^{1/p - 1/n} \mathcal{E}_p g. \tag{2}$$

The ordinary norms satisfy

$$\|f_\varepsilon\|_q = \varepsilon^{1/q} \|g\|_q \tag{3}$$

and, using only the first partial derivative,

$$\|\nabla f_\varepsilon\|_p \ge \|\partial_1 f_\varepsilon\|_p = \varepsilon^{1/p - 1} \|\partial_1 g\|_p. \tag{4}$$

Combining (2)–(4) gives

$$\begin{aligned} \frac{\mathcal{E}_p f_\varepsilon}{\|f_\varepsilon\|_q^{(n-1)/n} \|\nabla f_\varepsilon\|_p^{1/n}} &\le C_g \, \varepsilon^{1/p - 1/n - (n-1)/(nq) - (1/p - 1)/n} \\ &= C_g \, \varepsilon^{\frac{n-1}{n}(1/p - 1/q)}. \end{aligned}$$

The last exponent is strictly positive because $q > p$. Letting $\varepsilon \downarrow 0$ proves the theorem. $\square$

5 Checks, sharpness, and novelty status

The argument uses identities rather than asymptotic equivalences. It also covers $p = 1$: then the compressed first derivative has constant L^1 norm, while the final exponent remains positive for every $q > 1$. Because the bump is smooth, no approximation of characteristic functions or BV relaxation is needed.

The accompanying script checks the exponent simplification exactly with rational arithmetic and evaluates four representative parameter triples:

```
python code/verify_scaling.py
```

All checks pass. The script verifies algebra, not the analytic proof.

A bounded novelty search on 12 August 2026 used arXiv:2003.07391, the exact title, and combinations of "reverse Zhang", "affine Poincare", "q¿p", and "affine energy". It found the source and adjacent later affine Sobolev work, but no explicit resolution of item (6). Novelty confidence is moderate, pending a broader citation review.

Human-review recommendation. Verify the directional-unit-ball transformation and the determinant exponent in (1). Once that identity is confirmed, the remaining norm scalings and the contradiction are immediate.

References

[1] J. Haddad, C. H. Jiménez, and M. Montenegro, *From affine Poincaré inequalities to affine spectral inequalities*, arXiv:2003.07391 (2020), Section 7, item (6), PDF p. 27.